IBM watsonx Challenge 2026 · Growth Enablers Track · Sales Committee

ISC Automated Sales Forecast

AI-powered GM meeting prep — built entirely with IBM Bob

v2.6.3-rc1 · Path 1: Built During Challenge

BUILT WITH IBM BOB

IBM Bob (IBM's watsonx AI development assistant) was the **co-developer for every session** — code, architecture, debugging, documentation, and this document.

51	~100	Jul 10–21
Sessions with Bob	Hours w/ Bob	Built during challenge

PERSONAL & TEAM PRODUCTIVITY IMPACT

90 min	<5 min	94%	9
Manual weekly prep — before	Automated weekly prep — after	Time reduction, every week	IBM users — same time savings each

Every week, US Public Sector Sales VP **Dushyant K. Patel** and his team spent 60–90 minutes on manual GM meeting prep: exporting ISC pipeline data, hand-scoring deals, writing a narrative from memory, building a PowerPoint. **ISC Automated Sales Forecast eliminates that manual work entirely.** The same preparation — scored, narrativized, diff'd, formatted — now takes under 5 minutes. For one VP: ~75 hours saved per year. Across 9 IBM users on this team: ~675 hours per year. Multiplied across IBM's US Public Sector sales organization, the impact scales to thousands of hours annually.

THE PROBLEM IBM BOB HELPED SOLVE

Seven technical approaches failed. IBM Sales Cloud blocks all programmatic access via IBM w3id SSO with passkey authentication. SAQL API, Playwright automation, browser extensions, Salesforce Connected App — all dead ends.

IBM Bob worked through every approach with the team, helped diagnose each failure, and identified the HAR file as the viable path — capturing authenticated browser API responses with no credentials stored, no security risk.

51 sessions, one application. IBM Bob co-developed every layer: HAR parser, SQLite schema, watsonx.ai integration, diff engine, IBM Carbon UI, Pipeline Intelligence Engine, PPTX generator, Column Picker modal, Day 0 reset utility.

The complete session log — *UCC1-TechnicalSessionLog.md* — is in the repository. 51 sessions of unedited technical history: an IBMer and IBM Bob building a production application together.

WHAT IT DOES — FIVE STEPS, FIVE MINUTES

Refresh Data
IBM Bob built the HAR parser — loads full ISC pipeline in <2 sec, tagged per user. Replaces 20 min of ISC export + spreadsheet work.

Score with watsonx (ibm/granite-13b)
IBM Bob designed the scoring prompt and integration. Every deal scored 0–100 in ~15 sec. Replaces subjective eyeball scoring.

GM Narrative (meta-llama/llama-3-70b)
IBM Bob designed the narrative prompt and scoping logic. Meeting-ready prose generated in ~10 sec. Replaces writing from memory.

What Changed (ibm/granite-3-8b)
IBM Bob built the diff engine and delta summary. Stage changes, slippage, new deals — all surfaced automatically. Replaces manual comparison.

Custom PPT Column Picker
IBM Bob conceived and built the two-panel column picker modal. User selects columns, drags to reorder, generates IBM-branded PPTX in seconds.

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WHERE IBM BOB WAS USED — SESSION BY SESSION

AREA	WHAT IBM BOB DID	SESSIONS
Architecture design	HAR ingestion strategy (after 7 failed approaches), SQLite schema, multi-tenant user isolation model, baseline ledger design	1–5
IBM Carbon UI	Action bar, filter bar, named presets, tri-state checkbox, GM Ready indicator, sort modal, all CSS — IBM Design System applied throughout	6–15
watsonx.ai integration	Scoring prompts for granite-13b, narrative prompt for llama-3-70b, delta summary for granite-3-8b, mock fallback, streaming responses	10–20
Diff engine + audit ledger	computeDiff(), saveSnapshot(), confirmed_at < asOf guard, UUID-keyed append-only baseline_ledger, 34/34 test assertions	16–22
Pipeline Intelligence Engine	Hygiene scoring (0–100, 5 components), padded-close detection, manager-grouped rep dashboard, rep drill-down modal, coaching tooltips	23–40
PPTX generators	generatePpt() (GM Report), generatePtmpSlide() (PTMP format), generateCustomPpt() (Column Picker — dynamic widths, 12 fields, drag-to-reorder UI)	38–50
Multi-tenant SSO	Simulated IBM SSO, bcrypt auth, per-user data isolation, session architecture identical to real IBM w3id OIDC	25–30
Dev utilities + docs	seed-quarter.js, day0-reset.js (wiped 5,579 rows + 9,065 ledger rows, re-seeded 9 users in <2s), all HTML/MD documentation, this pitch document	48–51

IBM Bob quote from Session 1: "HAR ingestion turns IBM's own security posture into a feature — data is exported from an already-authenticated browser session, requiring no credentials stored in the application." That sentence became the foundation of the submission.

TECHNICAL ARCHITECTURE

Layer	Technology	IBM Bob's contribution
Frontend	Vanilla JS · IBM Carbon Design System	All UI components, modals, drag-to-reorder, CSS — every line
Server	Node.js 18 · Express · express-session	All endpoints, auth middleware, orchestration logic
Database	SQLite · better-sqlite3	Schema design, migrations, all queries
AI — Scoring	ibm/granite-13b-instruct-v2	Prompt design, integration, rule-based fallback
AI — Narrative	meta-llama/llama-3-70b-instruct	Prompt engineering, filter-scoping logic, streaming
AI — Delta	ibm/granite-3-8b-instruct	Diff summarization prompt, integration
PPT	pptxgenjs	All three generators: GM Report, PTMP, Column Picker
HAR Parser	Custom (scraper/load-from-har.js)	Full implementation — SAQL response extraction, field mapping

TEST COVERAGE — VALIDATES AI INTEGRATION RELIABILITY

34/34 assertions passing in scripts/test-baseline-ledger.js — a 13-week Q3 2026 in-memory simulation that IBM Bob wrote and debugged: baseline writes, diff computation, "today vs today" structural impossibility, quarter boundaries, multi-user isolation, confirmed_at guard. Passes in under 5 seconds on any Node.js 18+ machine. Run it: `node scripts/test-baseline-ledger.js`

IBM Bob — 51 sessions · ibm/granite-13b-instruct-v2 · meta-llama/llama-3-70b-instruct · ibm/granite-3-8b-instruct · IBM watsonx.ai · IBM Carbon Design System · Node.js 18 · Express · SQLite · github.com/jefftrbo/UCC1-ISCAutomatedSalesForecast · v2.6.3-rc1

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TEAM — BUILT WITH IBM BOB FROM DAY ONE

Jeffrey L. Trbovich — Technical Lead, IBM ATL / US Public Sector
Drove product vision, architecture decisions, and all 51 sessions with IBM Bob. Domain expert on IBM Sales VP workflows.

Dushyant K. Patel — Business Lead, Sales VP US Public Sector IBM
Business owner and primary user. Provided live ISC pipeline data (969 real opportunities) throughout development. Validated every feature against real weekly GM prep workflow.

IBM Bob — Co-developer, all 51 sessions
IBM's watsonx-powered AI development assistant. Co-wrote every line of production code, designed every architecture decision, debugged every issue, wrote all documentation including this pitch. The repository session log is the unedited proof.

Challenge Team
John Albertson · Kim Salatino · Justin Griffin · Kerry Hovan · Spencer Korn · Jeff Underwood · Muhammad Safwat — test users and challenge team members.

DEVELOPMENT TIMELINE — 12 DAYS WITH IBM BOB

Jul 10 Kick-off. Bob identified HAR ingestion as the only viable path past IBM w3id auth. SQLite schema. First PPTX. Sessions 1–3.

Jul 11–13 IBM Carbon Design System UI. Bob built action bar, named presets, diff engine, permanent audit ledger, 34/34 test harness. v2.2.0–v2.3.0.

Jul 14–15 Bob built simulated IBM SSO, multi-tenant isolation, 9 test users, per-user seed data, login page, header identity chip. v2.4.0.

Jul 16–18 Bob built Pipeline Intelligence Engine — hygiene scoring, Deal Health Card, padded-close detection, manager-grouped Rep Hygiene, rep drill-down. 21 micro-versions. v2.5.0–v2.5.21.

Jul 18–19 Bob built PTMP slide generator, multi-column sort modal, narrative hotfix. v2.6.0–v2.6.2-rc2 tagged as submission candidate.

Jul 21 Bob built Column Picker Custom PPT, Day 0 reset utility (wiped 5,579 rows, re-seeded 9 users in <2s), full doc sweep. v2.6.3-rc1 — final RC.

VERSION SUMMARY

Version	What Bob + Jeff shipped
v1.0–v2.1	HAR ingestion · SQLite · rule-based scoring · first PPTX — all built with Bob
v2.2–v2.3	IBM Carbon UI · diff engine · permanent audit ledger · 34/34 tests — all built with Bob
v2.4	Simulated IBM SSO · multi-tenant isolation · 9 users — built with Bob
v2.5.0–v2.5.21	Pipeline Intelligence Engine · Deal Health Card · Rep Hygiene coaching — built with Bob
v2.6.0–v2.6.2	PTMP generator · multi-column sort · narrative hotfix — built with Bob
v2.6.3-rc1	**Column Picker Custom PPT · Day 0 reset · final RC — built with Bob**

PATH TO PRODUCTION

The **CIO "Build with watsonx" Path to Production** is the identified deployment channel. A ServiceNow AI System Demand is being submitted with Dushyant Patel as Business Process Owner. Upon Technical Review Board approval: IBM Cloud Code Engine, CIO-managed watsonx.ai credentials, real IBM w3id SSO — available to any IBM Sales VP from any browser, zero installation.

Approval creates standing for a **Salesforce Connected App with OAuth 2.0**, eliminating the HAR step entirely. The same application, unchanged, serves any IBM Sales VP with a Salesforce-backed pipeline. **Client Zero: IBM using IBM Bob and watsonx to fix IBM's own workflow first.**

Repository: github.com/jefftrbo/UCC1-ISCAutomatedSalesForecast · Tag: v2.6.3-rc1 · 75+ commits
Session log: UCC1-TechnicalSessionLog.md — 51 sessions, complete IBM Bob co-development history
Test harness: node scripts/test-baseline-ledger.js — 34/34 assertions <5s · **Reset:** node scripts/day0-reset.js